BIOS 6602

Lecture 12

Log Transformations & Polynomial Regression

KKNM Chapter 15

{ LECTURE NOTE VERSION}

Data Transformations to Address Non-Linearity

- Linear regression methods can be used to model curves as long as those curves can be expressed in a linear fashion. The following are examples of curvilinear relationships that can be estimated using linear regression models:

  - $Y = e^{\beta_0} e^{\beta_1 X} e^{\varepsilon}$ → $\log(Y) = \beta_0 + \beta_1 X + E$ {log-transform *Y*}
  - $Y = \sqrt{\beta_0 + \beta_1 X + \varepsilon}$ → $Y^2 = \beta_0 + \beta_1 X + E$ {square *Y*}
  - $Y = \beta_0 + \beta_1 \log(X) + \varepsilon$ {log-transform *X*}
  - $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$ {add quadratic *X* term}
  - $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$ {add quadratic and cubic *X* terms}

- Transformations of the <u>independent</u> variables (i.e., predictors) are usually performed to address non-linearity (or to reduce leverage/influence), not to address non-normality since the independent variables need not be normally distributed. In fact, we have already seen the use of categorical variables as independent variables, which are far from normally distributed.

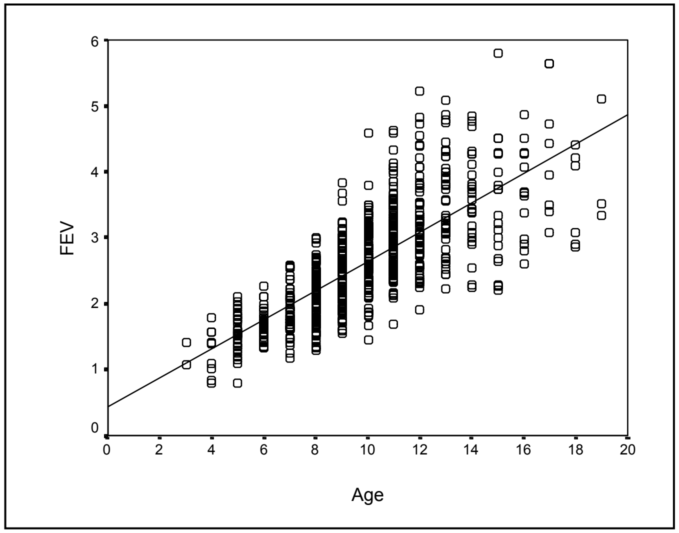
Data Transformations to Remove Heteroscedasticity

- The assumption of homoscedasticity is important, especially if the regression analysis is used for predictions.
- Transformations of the response variable (the dependent variable) are often used to remove heteroscedasticity. This type of transformation is called a ***variance-stabilization transformation***.
- Taking the logarithm of the response variable is a particularly useful transformation, especially for removing heteroscedasticity when the residual variance is an increasing function of X.
- Other transformations are sometimes used to stabilize the variance, but they may give a model that is more difficult to interpret.

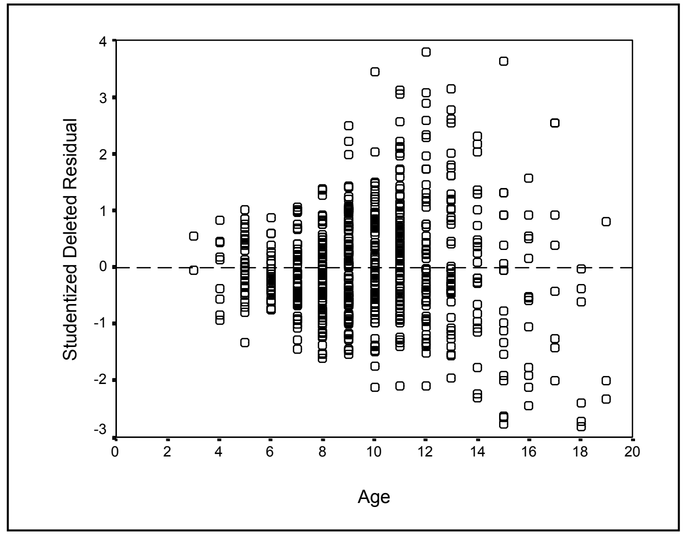
Relationship of σ^2 to $E[y]$	Transformation	Comment
$\sigma^2 \propto E[y]$	$\sqrt{y}$	Used for Poisson data
$\sigma^2 \propto E[y](1-E[y])$	$\sin^{-1}\sqrt{y}$	Used for binomial proportions or rates
$\sigma^2 \propto (E[y])^2$	$\log(y)$	Also used for non-linearity, non-normality; y>0
$\sigma^2 \propto (E[y])^3$	$y^{-1/2}$	
$\sigma^2 \propto (E[y])^4$	y^{-1}	

EXAMPLE: Residual Plots for Non-transformed FEV

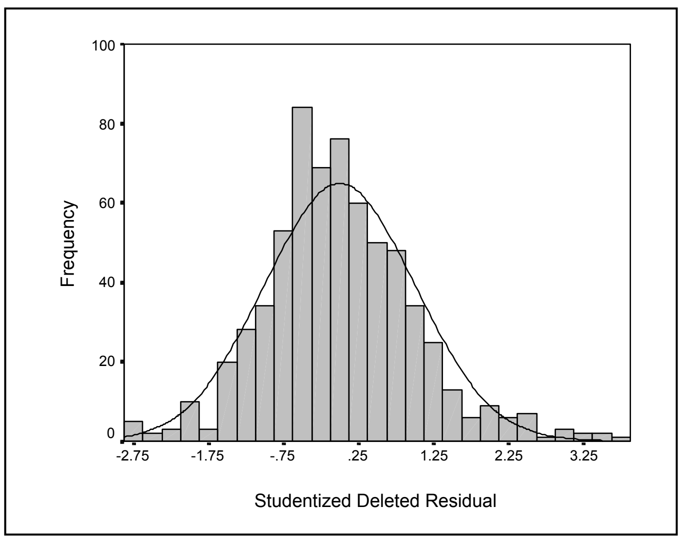
Scatterplot



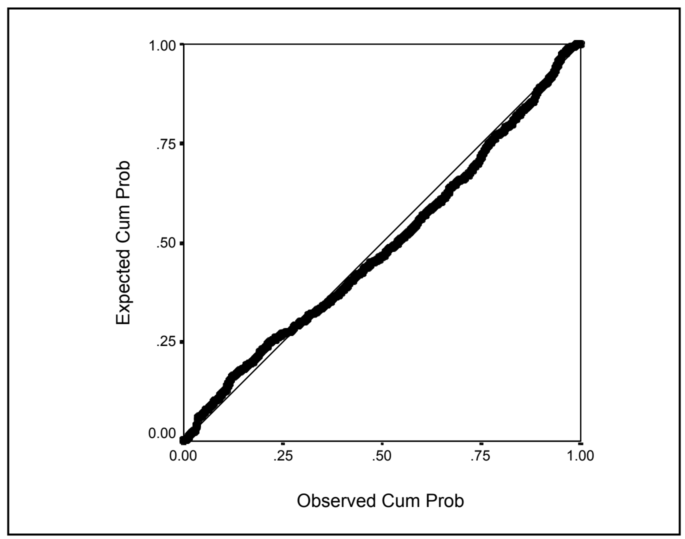
Residual Plot



Histogram

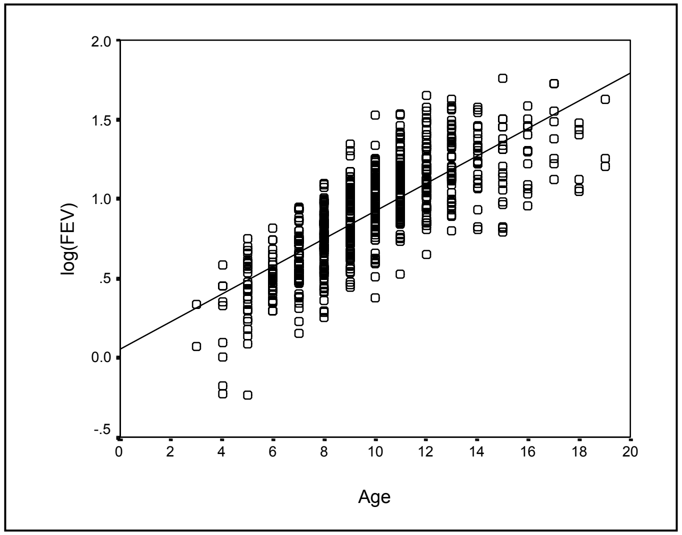


Normal Probability Plot

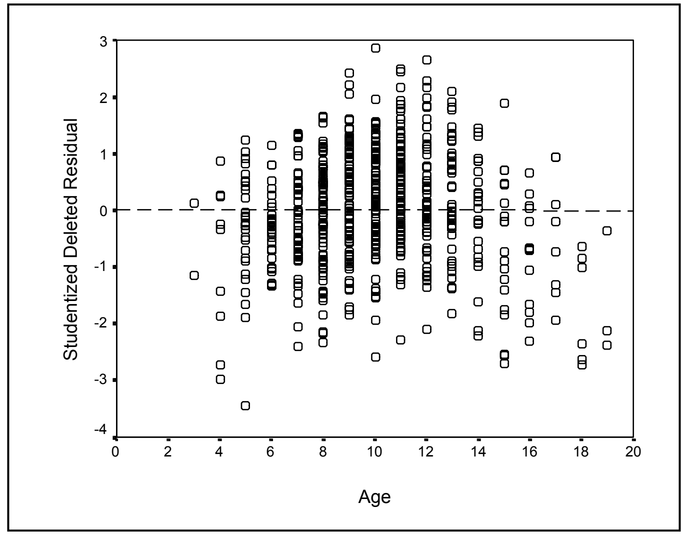


Residual Plots for Log-transformed FEV

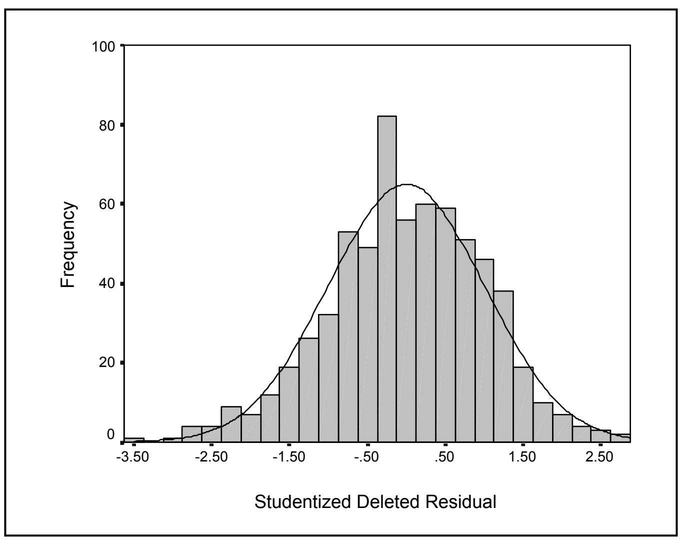
Scatterplot



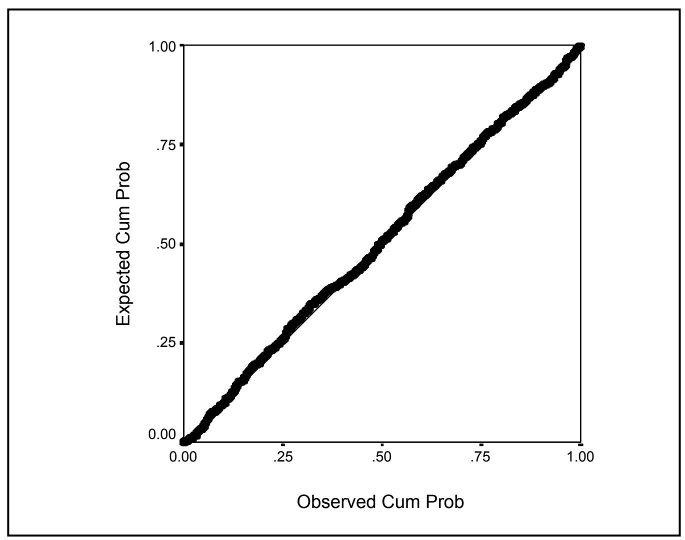
Residual Plot



Histogram



Normal Probability Plot



```
*** Straight-Line ***;
PROC REG DATA=fev;
 MODEL fev = age;
RUN;
```

Dependent Variable: fev FEV

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	280.91916	280.91916	872.18	<.0001
Error	652	210.00068	0.32209		
Corrected Total	653	490.91984			

Root MSE	0.56753	R-Square	0.5722
Dependent Mean	2.63678	Adj R-Sq	0.5716

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	0.43165	0.07790	5.54	<.0001
age	Age	1	0.22204	0.00752	29.53	<.0001

```
*** Log transformed FEV ***;
PROC REG DATA=fev;
 MODEL log_fev = age;
RUN;
```

Dependent Variable: log_fev LN(fev)

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	43.21005	43.21005	961.01	<.0001
Error	652	29.31586	0.04496		
Corrected Total	653	72.52591			

Root MSE	0.21204	R-Square	0.5958
Dependent Mean	0.91544	Adj R-Sq	0.5952

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	0.05060	0.02910	1.74	0.0826
age	Age	1	0.08708	0.00281	31.00	<.0001

There is a statistically significant association between age and FEV (p <.0001). Now we need to interpret the association!

Interpretation of Model with Log-Transformed Outcome
(Continuous Predictor)

When a logarithmic transformation of the dependent variable is used, the model is interpreted in a multiplicative scale.

Dependent: log(FEV)
Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	Intercept	1	0.05060	0.02910	1.74	0.0826	-0.00655	0.10774
age	Age	1	0.08708	0.00281	31.00	<.0001	0.08157	0.09260

- Transformed Interpretation:

$$\text{LN(FEV)} = 0.05060 + 0.08708 \times age$$

Intercept: At an age of 0, the predicted LN(FEV) is 0.05060.

Slope: For each year of age, LN(FEV) increases, on average, by 0.08708 units.

- Back-transformed: $\text{LN(FEV)} = 0.05060 + 0.08708 \times age$

$$\text{FEV} = e^{0.0506 + 0.0871 \times age} = e^{0.0506}\, e^{0.0871 \times age}$$

$$\text{FEV} = 1.052 \times (1.091)^{age}$$

Thus, for each year of age, FEV increases, on average, $e^{0.0871} = 1.091$ times (or ~9% per year).

- The 95% CI for the slope $[0.0871 \pm 1.96 \times 0.0028 = (0.082, 0.093)]$ can also be back-transformed:

$$(e^{0.082}, e^{0.093}) = (1.085, 1.097)$$

- What is the expected FEV for a 5-year old?

$\text{FEV} = 1.052 \times (1.091)^5$

$\text{FEV} = 1.052 \times 1.091 \times 1.091 \times 1.091 \times 1.091 \times 1.091 = 1.626$ L

~ or ~

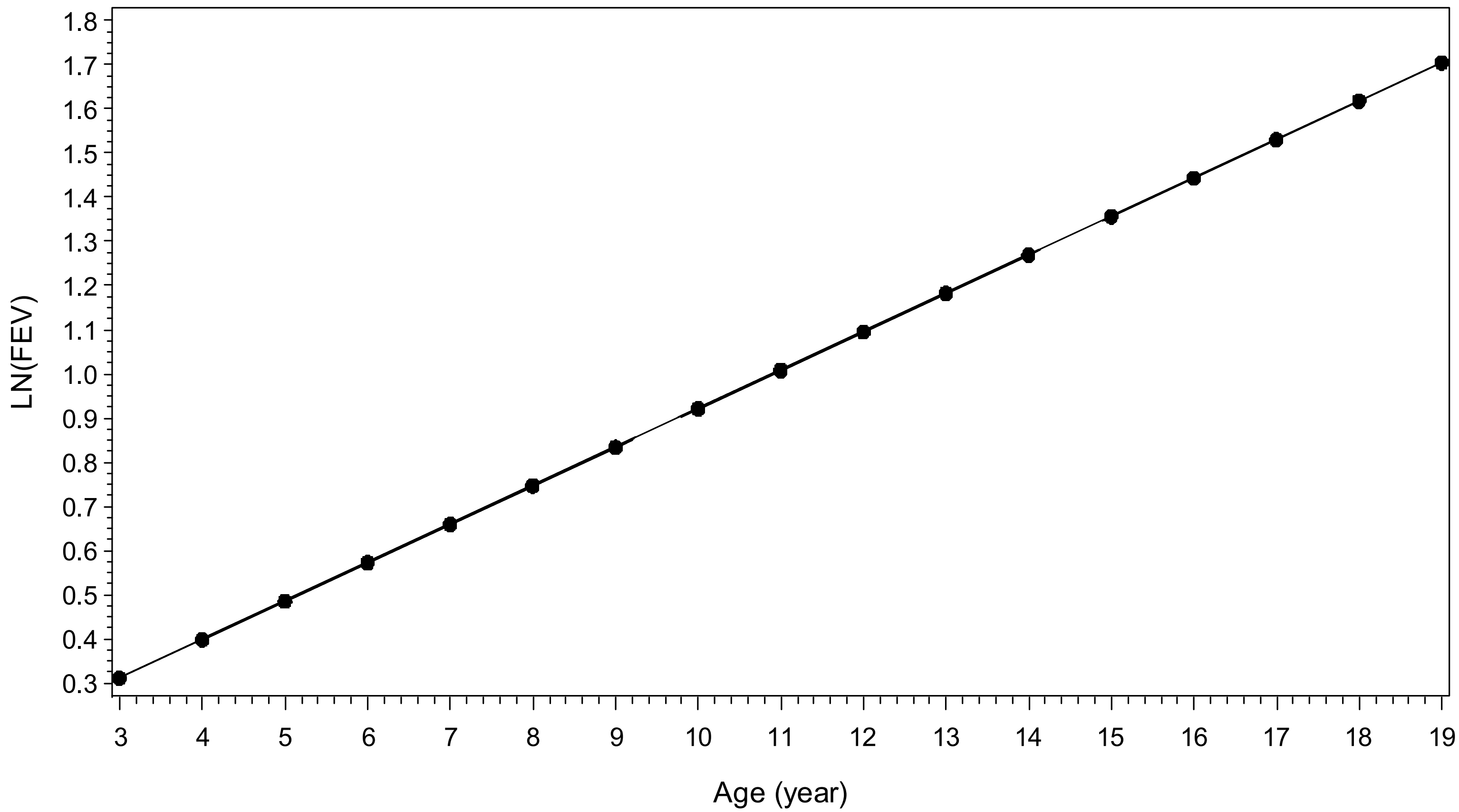
$\text{LN(FEV)} = 0.05060 + 0.08708 \times 5 = 0.486$

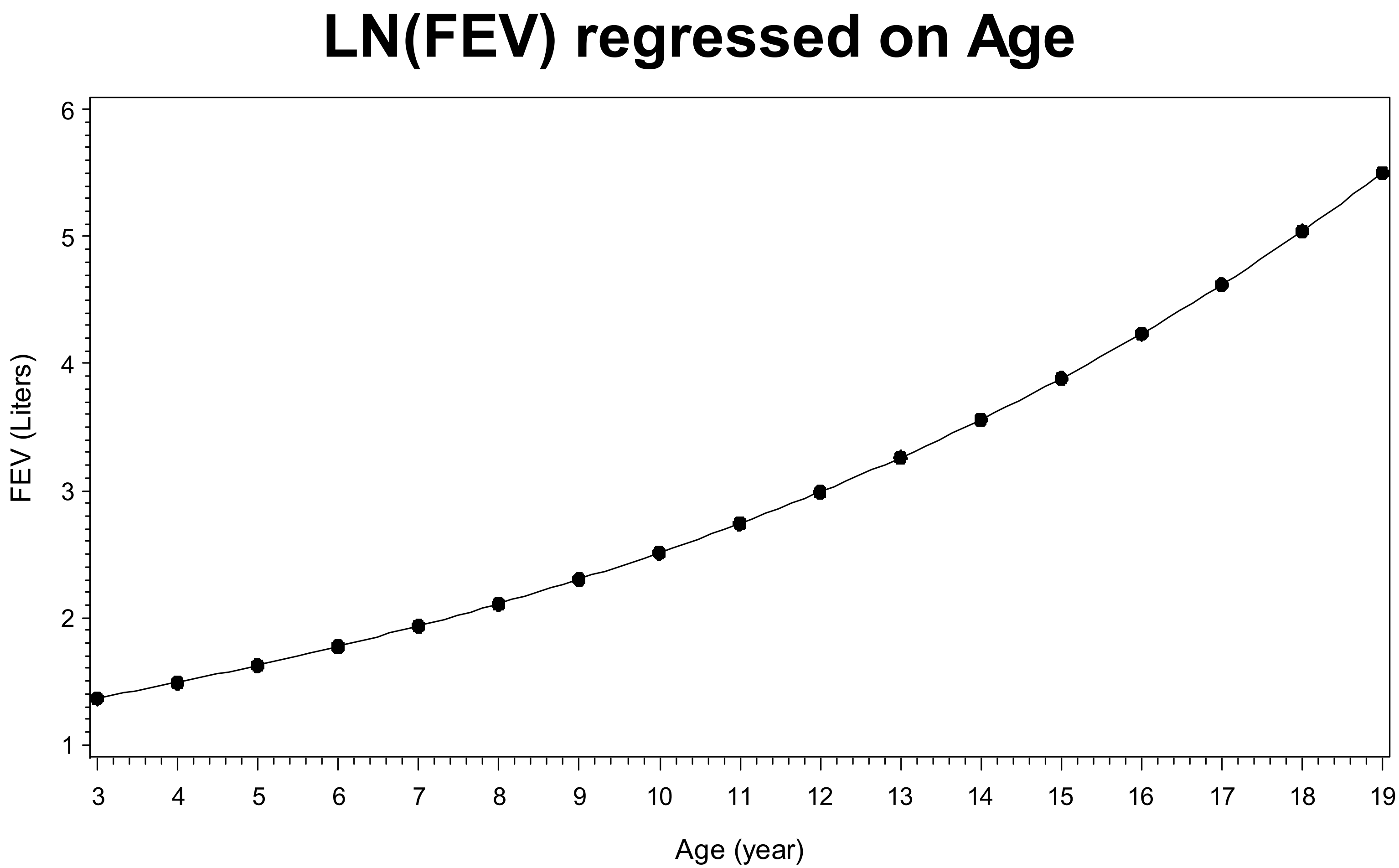
$\text{FEV} = \text{EXP}(0.486) = 1.626$ L

Examples	Absolute Change (Liters)	% Change
Age 5 to 6	0.14792 L	9.0984%
Age 10 to 11	0.22863 L	9.0984%
Age 15 to 16	0.35336 L	9.0984%

age	log_fev	fev
0	0.05060	1.05190
1	0.13768	1.14761
2	0.22476	1.25202
3	0.31184	1.36594
4	0.39892	1.49021
5	0.48600	1.62580
6	0.57308	1.77372
7	0.66016	1.93510
8	0.74724	2.11117
9	0.83432	2.30325
10	0.92140	2.51281
11	1.00848	2.74143
12	1.09556	2.99086
13	1.18264	3.26298
14	1.26972	3.55986
15	1.35680	3.88375
16	1.44388	4.23710
17	1.53096	4.62261
18	1.61804	5.04320

LN(FEV) Regressed on Age





Examples	Absolute Change (Liters)	% Change
Age 5 to 6	0.14792 L	9.0984%
Age 10 to 11	0.22863 L	9.0984%
Age 15 to 16	0.35336 L	9.0984%

Interpretation of Model with Log-Transformed Outcome (Binary Predictor)

```
PROC REG;
 MODEL logfev=csmoke /*0=Non-smoker;1=smoker*/;
 WHERE age ge 14;
RUN;
```

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	1.32664	0.03337	39.76	<.0001
csmoke	Current Smoker	1	-0.13022	0.05241	-2.48	0.0153

$$\textbf{Log(FEV)} = \mathbf{1.32664 - 0.13022} \times \boldsymbol{csmoke}$$

- Interpretation of the Intercept and Slope?

Intercept = 1.32664: Log(FEV) for non-smokers

Slope = -0.13022: difference in Log(FEV) between smokers and non-smokers.

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	1.32664	0.03337	39.76	<.0001
csmoke	Current Smoker	1	-0.13022	0.05241	-2.48	0.0153

- We can back-transform:

$$\text{FEV} = e^{1.32664-0.13022\times csmoke} = e^{1.32664}\, e^{-0.13022\times csmoke}$$

Non-smokers: $e^{1.32664}$ = <u>3.768 Liters</u>
Smokers: $e^{1.32664}\, e^{-0.13022} = 3.768 \times 0.878 =$ <u>3.308 Liters</u>

These represent the ***geometric mean*** FEV for non-smokers and smokers between ages 14-19.

$$\text{FEV} = 3.768 \times (0.878)^{csmoke}$$ ↑ (1-0.878)×100 = 12.2 %

- There is a significant association between smoking status and FEV ($p = 0.0153$). On average, FEV is 0.878 times lower (12.2% lower) in smokers compared to non-smokers.

- 95% CI?

95% CI on log scale: $-0.13022 \pm 1.96(0.05241)$
$= (-0.2329, -0.0275)$

Now Back-Transform: $(e^{-0.2329}, e^{-0.0275}) = (0.792, 0.973)$

We are 95% confident that FEV is between 2.7% and 20.8% lower in smokers compared to non-smokers.

Log-Transformed Predictor (base 2)

```
*** Log-transformed Age (base 2) ***;
PROC REG DATA=fev;
 MODEL fev = log2_age;    /* log2_age = LOG2(age) */
RUN;
```

age	log2 age	fev hat
1	0.00000	-2.07060
2	1.00000	-0.61990
3	1.58496	0.22871
4	2.00000	0.83080
5	2.32193	1.29782
6	2.58496	1.67941
7	2.80735	2.00203
8	3.00000	2.28150
9	3.16993	2.52801
10	3.32193	2.74852
11	3.45943	2.94800
12	3.58496	3.13011
13	3.70044	3.29763
14	3.80735	3.45273
15	3.90689	3.59713
16	4.00000	3.73220
17	4.08746	3.85908
18	4.16993	3.97871

Dependent Variable: fev FEV

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	279.91712	279.91712	864.95	<.0001
Error	652	211.00272	0.32362		
Corrected Total	653	490.91984			

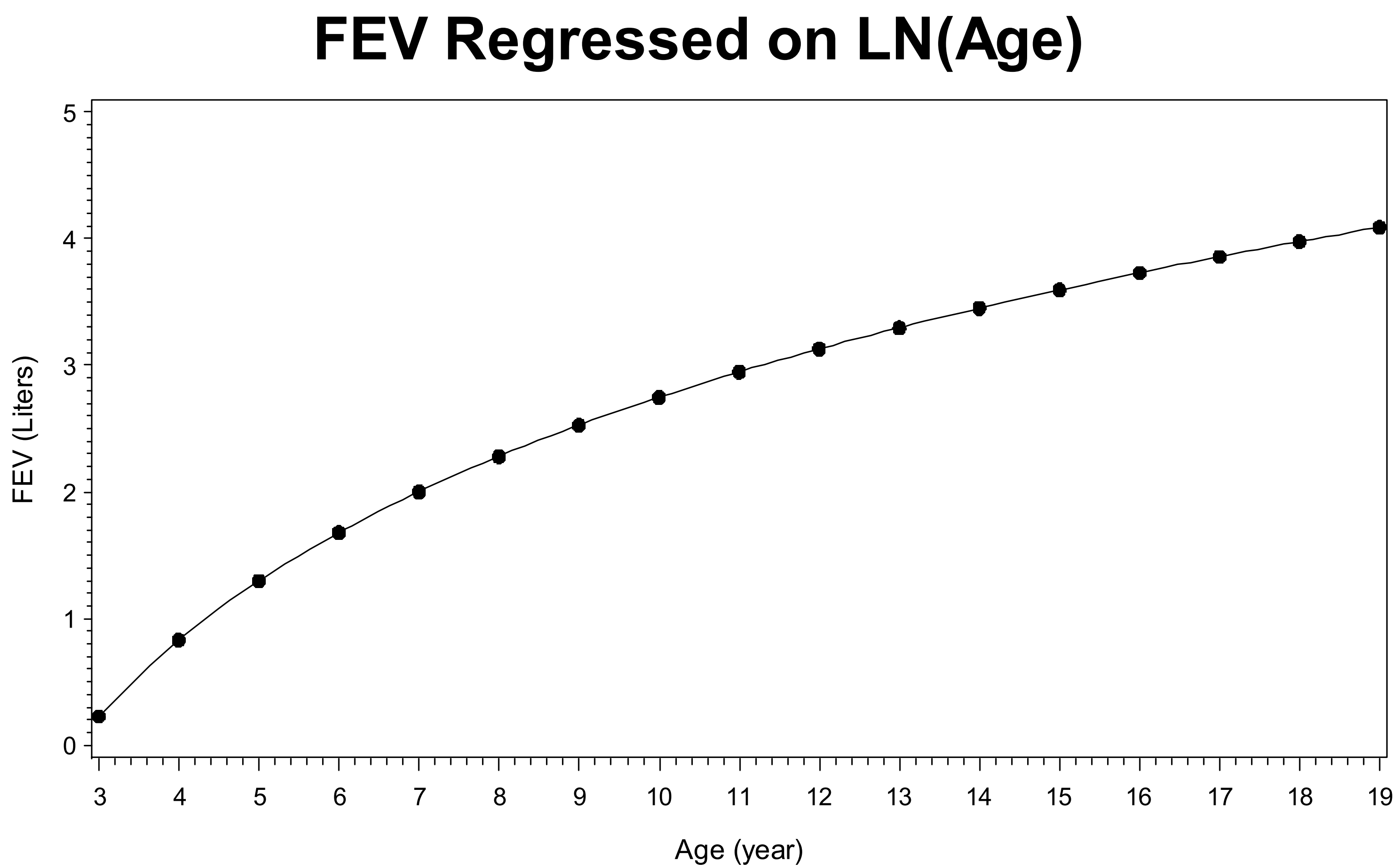
Root MSE	0.56888	R-Square	0.5702
Dependent Mean	2.63678	Adj R-Sq	0.5695

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	-2.07060	0.16160	-12.81	<.0001
log2_age		1	1.45070	0.04933	29.41	<.0001

Back-Transformed Interpretations

Intercept: Predicted FEV at age=1 year.

FEV increases by 1.45070 Liters for every doubling of age.


FEV Regressed on LN(Age)
FEV (Liters)
0
1
2
3
4
5
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Age (year)

Log-Transformed Predictor (natural log transform = base 2.71828)

```
*** Natural Log-transformed Age ***;
PROC REG DATA=fev;
 MODEL fev = log_age;    /* log_age = LOG(age) */
RUN;
```

Dependent Variable: fev FEV

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	279.91712	279.91712	864.95	<.0001
Error	652	211.00272	0.32362		
Corrected Total	653	490.91984			

Root MSE	0.56888	R-Square	0.5702
Dependent Mean	2.63678	Adj R-Sq	0.5695

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	-2.07060	0.16160	-12.81	<.0001
log_age		1	2.09292	0.07116	29.41	<.0001

These are the only two values that changed in this output when using a Natural Log transform instead of LOG-base 2.

<u>Back-Transformed Interpretations</u>

Intercept: Predicted FEV at age=1 year.

FEV increases by 2.09292 Liters for every **2.71828 times increase** in age.

Examples of calculations

Change in FEV for 2.71828 times increase in age:

Predicted FEV at Age = 5: -2.07060 + 2.09292*LOG(5) = 1.2978 L

Predicted FEV at Age = 13.5914: -2.07060 + 2.09292*LOG(13.5914) = 3.39074 L

Difference = 3.39074 - 1.2978 = 2.0929 ← Parameter estimate for model with Natural Log transformation.

Change in FEV for doubling in age:

Predicted FEV at Age = 5: -2.07060 + 2.09292*LOG(5) = 1.2978 L

Predicted FEV at Age = 10: -2.07060 + 2.09292*LOG(10) = 2.7485 L

Difference = 2.7485 - 1.2978 = 1.4507 ← Parameter estimate for model with Log (base 2) transformation.

Polynomial Models

A *k*th order polynomial in one variable, x, is an expression of the following form:

$$y = c_0 + c_1 x + c_2 x^2 + \ldots + c_k x^k$$

in which the c's and k (which must be a nonnegative whole number) are constants.

The statistical model is an expression of the following form:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \ldots + \beta_k X^k + E$$

This statistical model is a linear regression model because Y is a linear function of $\boldsymbol{\beta}$.

Polynomial models are useful:

- In situations where the analyst knows that curvilinear effects are present in the true response function.
- As approximating functions to unknown and possibly very complex nonlinear relationships.

Important considerations when using polynomial models include:

- Selecting the order of the model (model selection strategy)
- Extrapolation
- Ill-conditioning

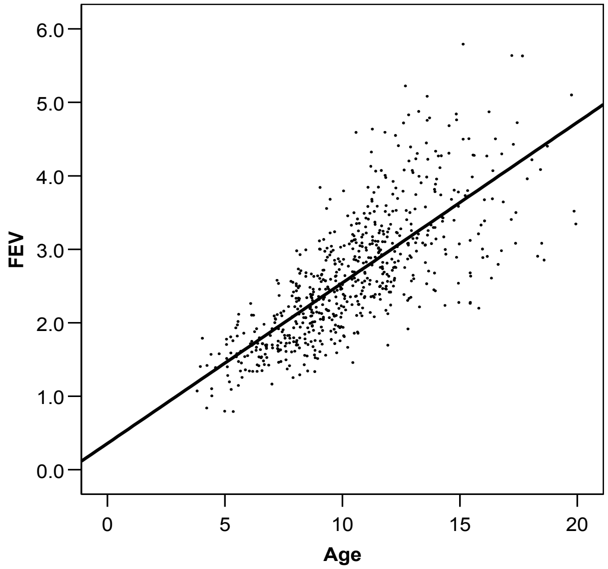
Fitting and testing Higher-Order Models

- How large an order of polynomial model to consider depends on the problem being studied and the amount and type of data being collected.
- A large number of well-placed predictor values and a small error variance are needed to obtain reliable fits for models of higher than order 3 (cubic).
- Fitting polynomial models or orders higher than three usually leads to models that are neither always decreasing nor always increasing. Substantial theoretical and/or empirical evidence should exist to support the employment of such complicated non-monotonic models.
- The quantity of data directly limits the maximum order of a polynomial that may be fit. Generally, the maximum-order polynomial that may be fit is one less than the number of distinct X-values (a polynomial curve of order d-1 can pass exactly through d distinct X-values).
- Collinearity problems can arise in polynomial models (centering the predictors by subtracting the mean of each variable from each of the observed values on that variable, can help remedy such problems for the second-order polynomial model).
- In general, lower-order terms are not removed from a model if higher-order terms are significant. Removal of the 1st order term corresponds to the hypothesis that the predicted response is symmetric about x=0 and has an optimum (minimum or maximum) at x=0. Often this hypothesis is not meaningful and should not be considered. Only when this hypothesis is scientifically justifiable should we consider removing the lower order term.

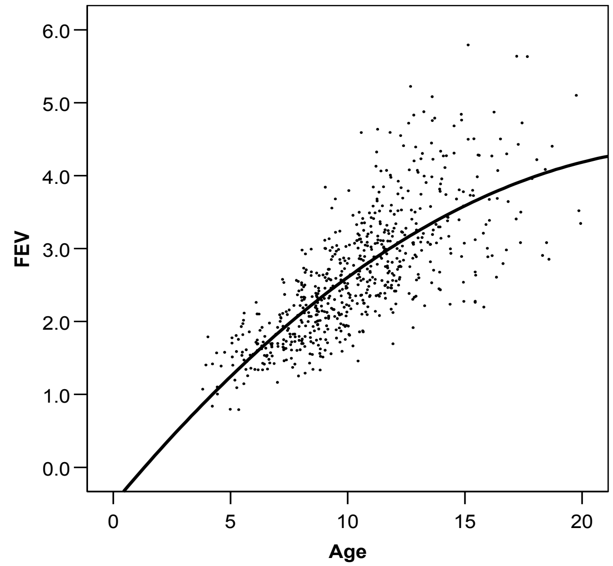
  Note: A similar argument can be made about removing β_0 from a model. This would assume the predicted value at X=0 is Y-hat=0.

Polynomial Models for FEV data

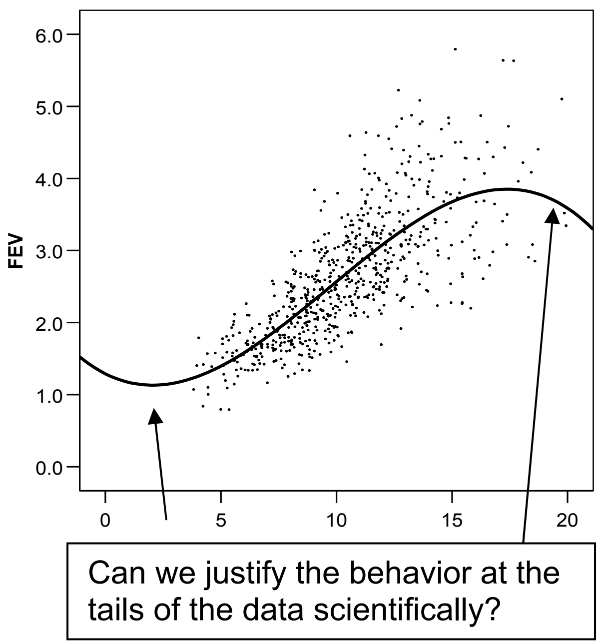
Straight Line Model:



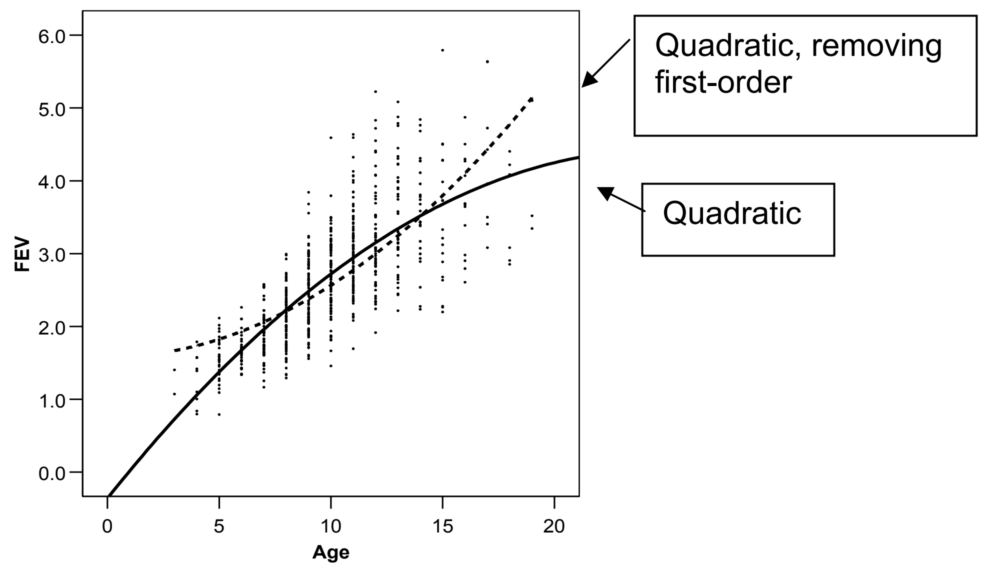
Quadratic Polynomial:



Cubic Polynomial:



Quadratic polynomial & Quadratic removing first order:



Linear Model ($R^2 = 0.5722$) Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	280.91916	280.91916	872.18	<.0001
Error	652	210.00068	0.32209		
Corrected Total	653	490.91984			

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	0.43165	0.07790	5.54	<.0001
age	Age	1	0.22204	0.00752	29.53	<.0001

Quadratic Model ($R^2 = 0.5840$) Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	286.70559	143.35280	456.98	<.0001
Error	651	204.21424	0.31369		
Corrected Total	653	490.91984			

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	-0.36036	0.19979	-1.80	0.0717
age	Age	1	0.38571	0.03882	9.94	<.0001
age2		1	-0.00776	0.00181	-4.29	<.0001

Cubic Model ($R^2 = 0.5925$)

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	290.87948	96.95983	315.06	<.0001
Error	650	200.04035	0.30775		
Corrected Total	653	490.91984			

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	1.23226	0.47558	2.59	0.0098
age	Age	1	-0.13324	0.14607	-0.91	0.3620
age2		1	0.04387	0.01413	3.10	0.0020
age3		1	-0.00159	0.00043051	-3.68	0.0002

Quartic Model ($R^2 = 0.5942$)

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	291.69429	72.92357	237.56	<.0001
Error	649	199.22555	0.30697		
Corrected Total	653	490.91984			

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	2.82562	1.08723	2.60	0.0096
age	Age	1	-0.85652	0.46730	-1.83	0.0673
age2		1	0.15722	0.07099	2.21	0.0271
age3		1	-0.00894	0.00453	-1.97	0.0491
age4		1	0.00016800	0.00010312	1.63	0.1038

Quadratic Model, removing first order polynomial (R^2 = 0.5209)

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	255.74206	255.74206	709.01	<.0001
Error	652	235.17777	0.36070		
Corrected Total	653	490.91984			

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	1.57789	0.04618	34.17	<.0001
age2		1	0.00986	0.00037048	26.63	<.0001

Best Model?

Cubic model, statistically. But does this model make sense scientifically (rapidly decreasing FEV after age 17)?

It is very difficult to interpret the coefficients from polynomial models. When fitting polynomial models, it is best to plot the regression equation!

Other Remedies for Non-Linearity

Sometimes we find that a low-order polynomial provides a poor fit to the data, and increasing the order of the polynomial or transforming *X* or *Y* doesn't help (the residual plot still exhibits some structure). One flexible approach is the use of **spline functions** to perform **piecewise polynomial fitting**.

- Divide the range of *X* into segments.
- Fit an appropriate curve of order *k* in each segment.

The piecewise linear spline (*k*=1) with a single knot (change point between segments) is given by:

$$E(Y) = \beta_{00} + \beta_{01}X + \beta_{10}(X-T)_+^0 + \beta_{11}(X-T)_+^1$$

$$(X-T)_+ = \begin{cases} (X-T) & \text{if } X\text{-}T > 0 \\ 0 & \text{if } X\text{-}T \le 0 \end{cases}$$

If X ≤ T, the straight-line model is

$$E(Y) = \beta_{00} + \beta_{01}X$$

And if X > T then the straight-line model is:

$$E(Y) = \beta_{00} + \beta_{01}X + \beta_{10} + \beta_{11}(X-T)$$

A smoother function would result if we required the regression function to be continuous at the knot. This can be accomplished by deleting the $\beta_{10}(X-T)_+^0$ term from the model above:

$$E(Y) = \beta_{00} + \beta_{01}X + \beta_{11}(X-T)_+^1$$

If X ≤ T, the straight-line model is

$$E(Y) = \beta_{00} + \beta_{01}X$$

And if X > T then the straight-line model is:

$$\begin{aligned} E(Y) &= \beta_{00} + \beta_{01}X + \beta_{11}(X-T) \\ &= (\beta_{00} - \beta_{11}T) + (\beta_{01} + \beta_{11})X \end{aligned}$$

- We usually assume that the positions of the knots are known. It they are parameters to be estimated, the resulting problem is a nonlinear regression problem.
- A **cubic spline** is a piecewise polynomial of order 3 (which is usually adequate for most practical problems).
- A potential disadvantage of cubic splines is that the **X'X** matrix can become ill-conditioned if there are a large number of knots. This problem can be overcome by using a different representation of the spline called the **cubic B-spline**.

```
*** Data Step Code for fitting piecewise polynomial models ***;
*** Allowing for a knot at age=14 ***;

I14 = (age ge 14);
age14 = age-14;
age14_I14 = I14*age14;

LABEL    I14       = 'I(age>=14)'
         age14     = '(Age-14)'
         age14_I14 = '(Age-14)*I(age>=14)';
```

Example: FEV Data (***Discontinuity at the Knot***):

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	290.84721	96.94907	314.97	<.0001
Error	650	200.07262	0.30780		
Corrected Total	653	490.91984			

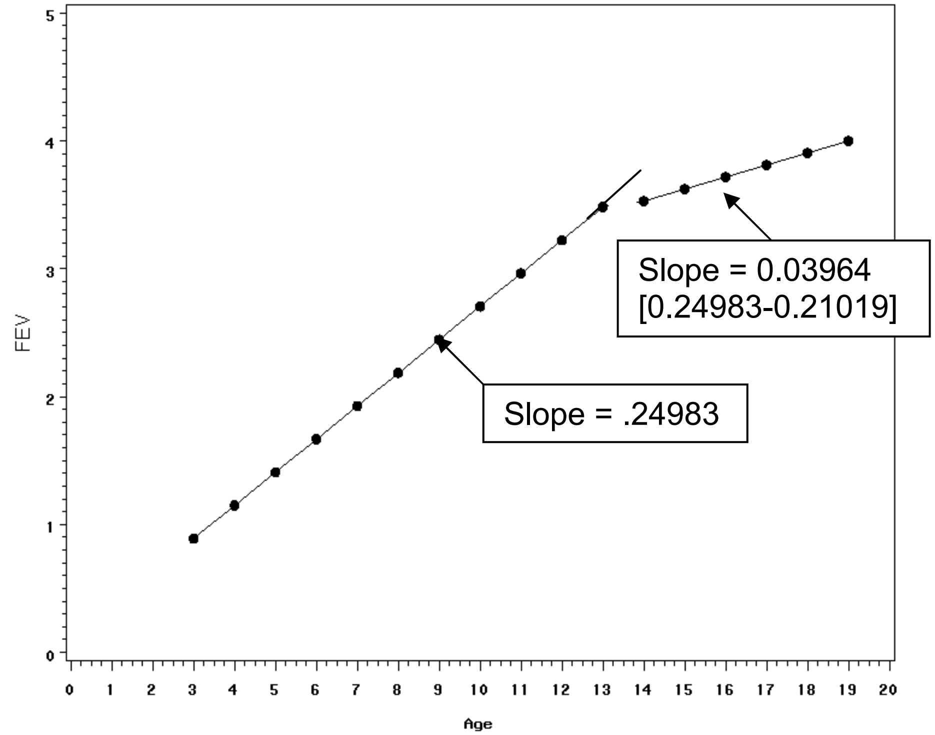
Parameter Estimates

	Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
β_{00}	Intercept	Intercept	1	0.11439	0.09638	1.19	0.2357
β_{01}	age	Age	1	0.25916	0.01014	25.55	<.0001
β_{10}	I14	I(age>=14)	1	-0.21310	0.10596	-2.01	0.0447
β_{11}	age14_I14	(Age-14)*I(age>=14)	1	-0.16499	0.04551	-3.63	0.0003

```
PROC REG DATA=fev;
 MODEL fev = age I14 age14_I14;
 OUTPUT out = pred2 predicted=p;
RUN;

PROC SORT DATA=pred2;
 BY age;
RUN;

PROC GPLOT DATA=pred2;
 plot p*age / VAXIS=axis1 HAXIS=axis2;
 SYMBOL INTERPOL=join VALUE=dot
COLOR=black;
RUN;
```



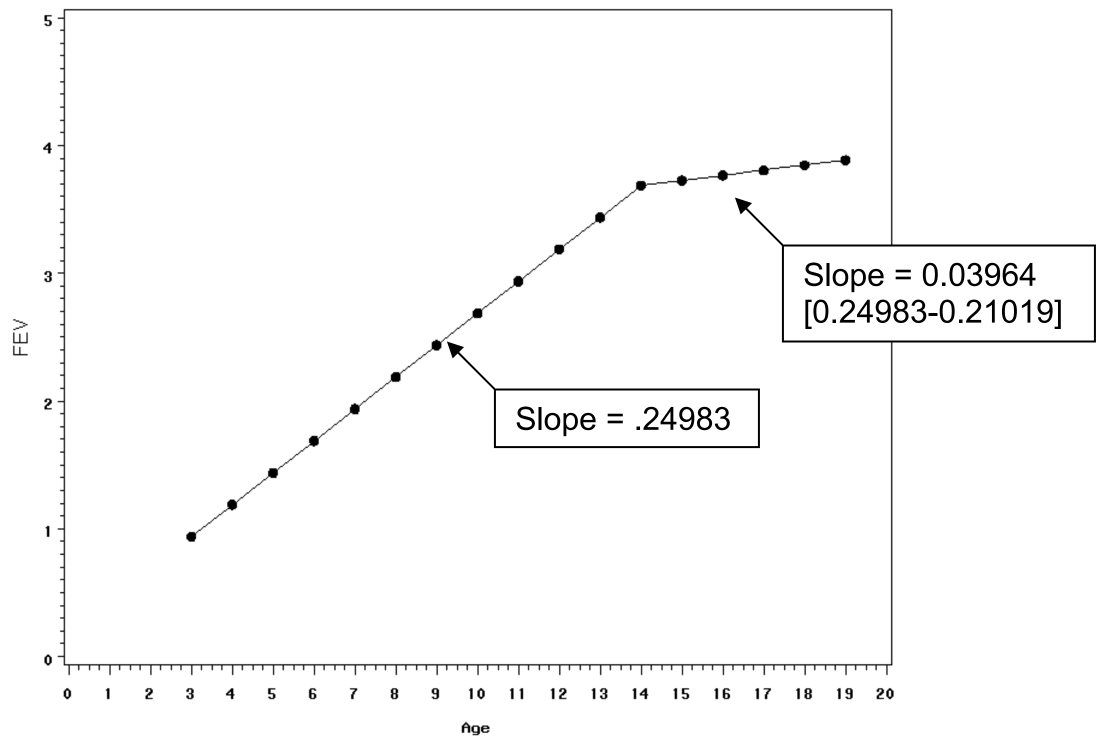
***Continuity at the Knot*:**

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	289.60233	144.80116	468.24	<.0001
Error	651	201.31751	0.30924		
Corrected Total	653	490.91984			

Parameter Estimates

	Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
β_{00}	Intercept	Intercept	1	0.19034	0.08888	2.14	0.0326
β_{01}	age	Age	1	0.24983	0.00904	27.63	<.0001
β_{11}	age14_I14	(Age-14)*I(age>=14)	1	-0.21019	0.03967	-5.30	<.0001




```
PROC REG DATA=fev;
 MODEL fev = age age14_I14;
 OUTPUT out = pred3 predicted=p;
RUN;
```

```
PROC SORT DATA=pred3;
 BY age;
RUN;

PROC GPLOT DATA=pred3;
 PLOT p*age / VAXIS=axis1 HAXIS=axis2;
 SYMBOL INTERPOL=join VALUE=dot COLOR=black;
RUN;
```

Nonparametric Methods

Nonparametric regression models are also available, including **Kernel** regression and **Loess,** both of which use data from a “neighborhood” around specific locations to estimate the regression line.

LOESS (locally weighted polynomial regression)

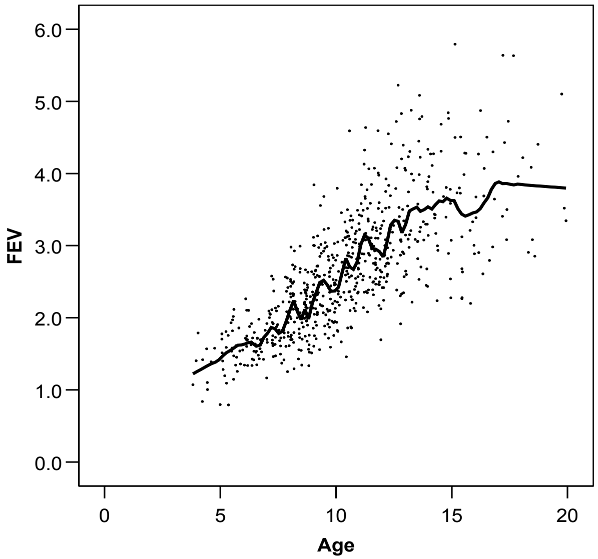
- At each point in the data set a low-degree polynomial is fit to a subset of the data, with explanatory variable values near the point whose response is being estimated.
- The polynomial is fit using weighted least squares, giving more weight to points near the point whose response is being estimated and less weight to points further away.
- The value of the regression function for the point is then obtained by evaluating the local polynomial using the explanatory variable values for that data point.
- The LOESS fit is complete after regression function values have been computed for each of the ***n*** data points.
- Many of the details of this method, such as the size of the neighborhood, degree of the polynomial model and the weights, are flexible can be chosen by the analyst.

```
*** LOESS REGRESSION ***;
PROC LOESS DATA=fev;
 MODEL fev=age / smooth=.99;
 ODS OUTPUT outputstatistics=stats;
RUN;
```

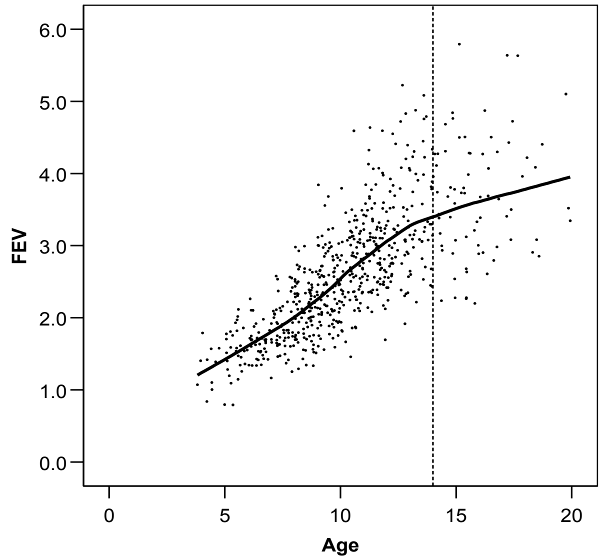
```
PROC SORT DATA=stats;
 BY age;

PROC GPLOT DATA=stats;
  PLOT (depvar pred)*age /overlay;
  symbol1 c=black i=rl value=dot width=2;
  symbol2 c=red i=join value=none width=2;
RUN;
```

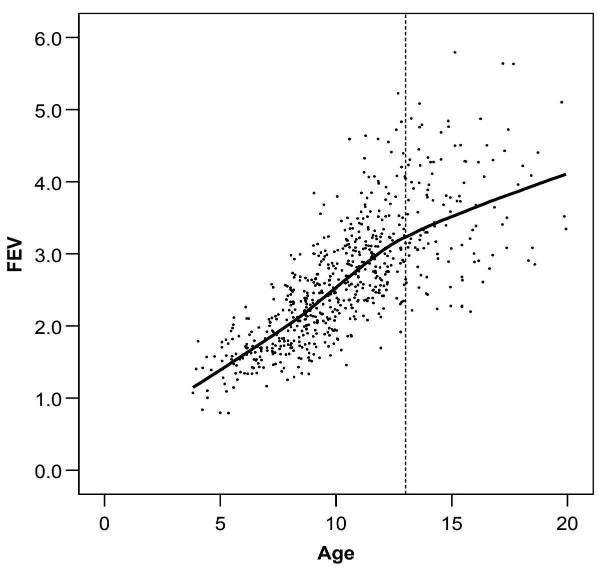
Loess: 5% of data points



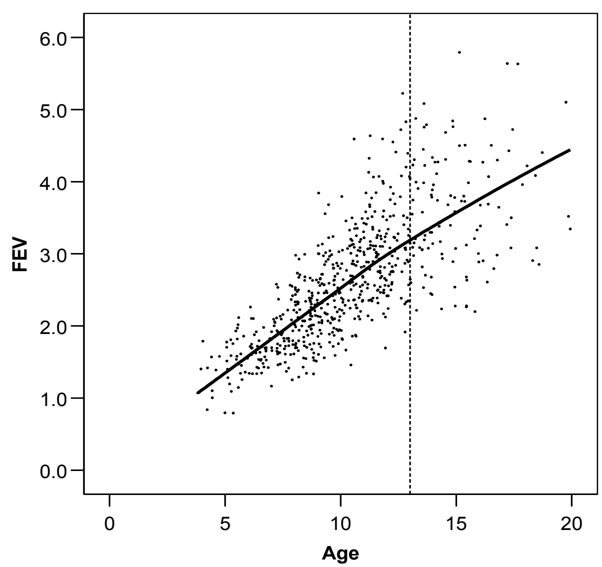
Loess: 50% of data points



Loess: 75% of data points



Loess: 99% of data points



THE REMAINING LECTURE NOTES ARE FYI ONLY

Appendix 1
The Geometric Mean

$$\text{Geometric Mean} = \left(\prod_{i=1}^{n} X_i\right)^{1/n} = \exp\left(\frac{\sum_{i=1}^{n}\log(x_i)}{n}\right)$$

$$\mathrm{e}\left(\frac{\sum_{i=1}^{n}\log(x_i)}{n}\right) = \mathrm{e}\left(\frac{1}{n}\sum_{i=1}^{n}\log(x_i)\right) = \mathrm{e}\left(\frac{1}{n}\log\left(\prod_{i=1}^{n} x_i\right)\right) = \mathrm{e}\left(\log\left(\prod_{i=1}^{n} x_i\right)\right)^{1/n} = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$$

Example: three data points: 2, 3, 36

Arithmetic Mean: "Balancing point on the arithmetic scale" $\sum_{i=1}^{n}\left(X_i - \bar{X}\right) = 0$

2 3 ($\bar{X}_{arithmetic}$ =13.66) 36

-10.66 +22.33

-11.66

= -22.33

Geometric Mean is "Balancing point on the multiplicative scale" $$\prod_{i=1}^{n} \frac{X_i}{\bar{X}_{geometric}} = 1$$

2 3 ($\bar{X}_{geometric} = 6$) 36

1/2 ×6

1/3

= 1/6

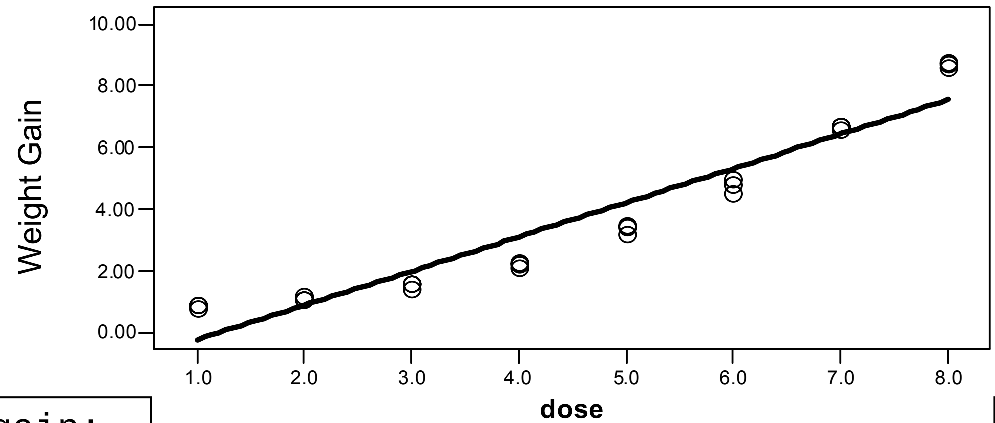
$$\left[2 \times 3 \times 36\right]^{1/3} = 6$$

$$e^{\left(\frac{\ln(2)+\ln(3)+\ln(6)}{3}\right)} = e^{1.7918} = 6$$

Example with Categorical Predictor

Example: (from KKNM) A laboratory study is undertaken to determine the relationship between the dosage (X) in milligrams of a certain drug and weight gain (Y) measured in ounces. 24 laboratory animals of the same sex, age, and size are selected and 3 animals are randomly assigned to each dose group.

Scatterplot:




```
PROC REG DATA=wtgain;
 MODEL wgtgain = dose;
RUN;
```

NOTE: X is fixed by the investigator in this example

The REG Procedure
Dependent Variable: wgtgain

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	155.66669	155.66669	238.27	<.0001
Error	22	14.37290	0.65331		
Corrected Total	23	170.03958			

MSE contains pure error + lack of fit error

Root MSE	0.80828	R-Square	0.9155
Dependent Mean	3.65417	Adj R-Sq	0.9116

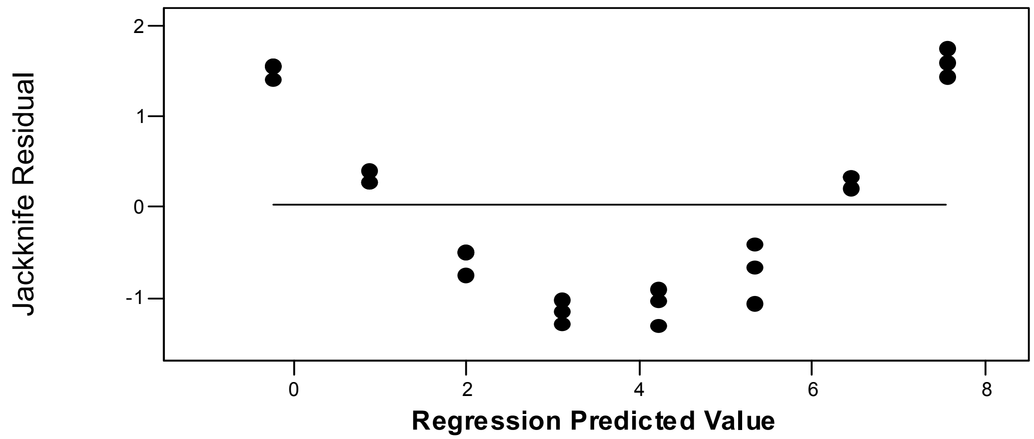
R-squared is fairly large, even though the relationship is not a straight-line.

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	-1.34762	0.36362	-3.71	0.0012
dose	1	1.11151	0.07201	15.44	<.0001

Intercept: Predicted weight gain when dose = 0. This is beyond the range of our data.
β1: For every one mg increase in dose, weight gain increases by an average of 1.11 ounces.

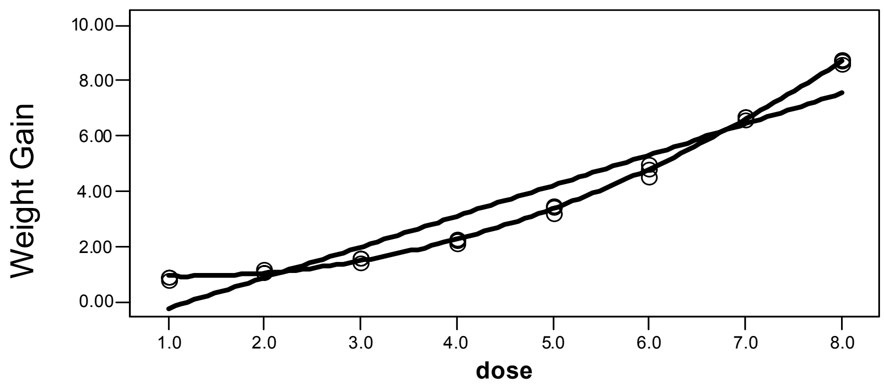
Residual Plot:

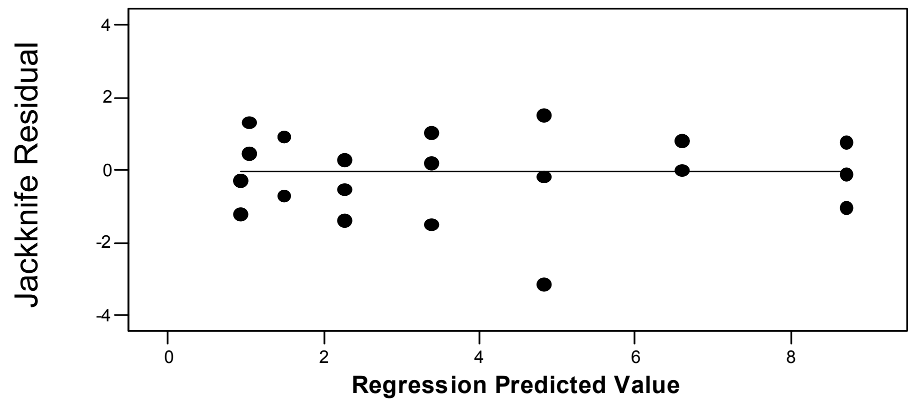


Does a straight-line model fit the data? **NO. There is an obvious pattern in the residuals.**

Is there a significant linear association between drug dose and weight loss?
Yes. On average, there is a 1.11 ounce increase per mg increase in dose. t = 15.44, p <0.001 { or F = 238.27}

What if we fit a polynomial model of order-2?






```
PROC REG DATA=wtgain;
 MODEL wgtgain = dose dosesq;
RUN;
```

dosesq = dose*dose

```
 The REG Procedure
Dependent Variable: wgtgain

Analysis of Variance
```

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	169.70004	84.85002	5247.78	<.0001
Error	21	0.33954	0.01617		
Corrected Total	23	170.03958			

MSE reduced dramatically
(SSE reduced, DF decreased)

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	1.15536	0.10242	11.28	<.0001
dose	1	-0.39028	0.05222	-7.47	<.0001
dosesq	1	0.16687	0.00566	29.46	<.0001

Note that b_0 and b_{dose} estimates changed when b_{dosesq} was added to the model. b_{dose} is now negative.

Covariance of Estimates

Variable	Intercept	dose	dosesq
Intercept	0.0104904359	-0.004908369	0.0004812127
dose	-0.004908369	0.0027268717	-0.000288728
dosesq	0.0004812127	-0.000288728	0.0000320808

Questions about the association between drug dose and weight loss for the quadratic model:

- Is the overall regression significant? That is, is more of the variation in Y explained by the second-order model than by ignoring X completely (and just using $\bar{Y}$).

<u>Overall F Test</u>

F = 5247.78   p <.0001

- Does the second-order model provide significantly more predictive power than the straight-line model does?

<u>Partial F test or t statistic for the beta coefficient for the quadratic term</u>

t = 29.46   p <.0001

- What is the R-square for the second-order model?

- Given that a second-order model is more appropriate than a straight-line model, should we add higher order terms to the second-order model?

Estimate of $\sigma^2_{Y|X}$ and Lack of Fit

- Given that a second-order model is more appropriate than a straight-line model, should we add higher order terms to the second-order model?

- Recall from previous lectures that we use ***MS$_{Error}$*** to estimate $\sigma^2_{Y|X}$. This estimate is given by the formula:

$$\hat{\sigma}^2_{Y|X} = \frac{\sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2}{n-2} = \frac{SS_{Error}}{n-2} = MS_{Error}$$

- The Mean Square Error (MS_{Error}) will only provide an unbiased estimate of the error variance when the hypothesized model is correct (in this case, if a straight-line model is appropriate), otherwise the MS_{Error} will estimate a quantity larger than the error variance. $\hat{\sigma}^2_{Y|X} > \sigma^2_{Y|X}$.
  - If the model is incorrect, then two factors contribute to the inflation of the SSE. The true variability in Y (the "***pure error***") and the error due to fitting an incorrect model (the "***lack of fit***" error)
  - With replicate observations (i.e., multiple observations with the same values of the predictor(s)), we can test for lack of fit of the assumed model by obtaining an estimate of $\sigma^2_{Y|X}$ that does not assume the correctness of the straight-line model (it is a model-free estimate of the residual variance or the "pure error").

Calculation of sums of squares due to ***pure error***:

Dose (X)	Weight Gain (Y)			$\bar{Y}_x$	SS_{PE} $\sum_m (Y_{mx} - \bar{Y}_x)^2$	df
1	.9	.9	.8	.866667	.006667	2
2	1.1	1.1	1.2	1.133333	.006667	2
3	1.6	1.6	1.4	1.533333	.026667	2
4	2.3	2.1	2.2	2.200000	.020000	2
5	3.5	3.4	3.2	3.366667	.046667	2
6	5.0	4.5	4.8	4.766667	.126667	2
7	6.6	6.7	6.7	6.666667	.006667	2
8	8.7	8.6	8.8	8.700000	.020000	2
					Σ=0.2600	**Σ=16**
				MS_{PE} = 0.2600/16 = 0.01625		

$\hat{\sigma}^2_{Y|X}$

This estimate $\hat{\sigma}^2_{Y|X}$ of is not model dependent. It is the "pure error"

So, to test the linear trend using the "pure error":

$$t = \frac{\hat{\beta}_{dose}}{\sqrt{\frac{MSE(pure)}{MSE(pure+LOF)} \times (SE(\hat{\beta}_{dose}))^2}} = \frac{1.11151}{\sqrt{\frac{0.01625}{0.65331} \times (0.072012)^2}} = \frac{1.11151}{\sqrt{0.02487 \times (0.072012)^2}} = 97.87$$

$$F = t^2 = 9578.54$$

Previous t=15.44
Variance was 40 times higher due to lack of fit error.

You can also obtain the "pure error" by fitting a "saturated model" – a model using a dummy code for each level *k* of X (or *k* -1 dummy codes if fitting an intercept as below)

```
                          The REG Procedure
                     Dependent Variable: wgtgain

                        Analysis of Variance

                                     Sum of           Mean
Source                  DF          Squares         Square    F Value    Pr > F

Model                    7        169.77958       24.25423    1492.57    <.0001
Error                   16          0.26000       0.01625
Corrected Total         23        170.03958
```

Pure Error

```
                         Parameter Estimates
                      Parameter       Standard
   Variable     DF     Estimate          Error     t Value     Pr > |t|

   Intercept     1      0.86667        0.07360       11.78       <.0001
   dose2         1      0.26667        0.10408        2.56       0.0209
   dose3         1      0.66667        0.10408        6.41       <.0001
   dose4         1      1.33333        0.10408       12.81       <.0001
   dose5         1      2.50000        0.10408       24.02       <.0001
   dose6         1      3.90000        0.10408       37.47       <.0001
   dose7         1      5.80000        0.10408       55.72       <.0001
   dose8         1      7.83333        0.10408       75.26       <.0001
```

The intercept is the predicted weight gain when dose2 – dose8 are zero, which occurs for dose group 1 (the reference group). Thus the intercept is the predicted weight gain (which is equal to the observed weight gain) for dose group 1.

SAS Coding:

```
dose1 = (dose=1);
dose2 = (dose=2);
dose3 = (dose=3);
dose4 = (dose=4);
dose5 = (dose=5);
dose6 = (dose=6);
dose7 = (dose=7);
dose8 = (dose=8);
```

Dose Group	Variable Coding for Model						
	dose2	dose3	dose4	dose5	dose6	dose7	dose8
1	0	0	0	0	0	0	0
2	1	0	0	0	0	0	0
3	0	1	0	0	0	0	0
4	0	0	1	0	0	0	0
5	0	0	0	1	0	0	0
6	0	0	0	0	1	0	0
7	0	0	0	0	0	1	0
8	0	0	0	0	0	0	1

- The difference in the Regression Sum of Squares between the lower-order model being considered and the full model containing all higher-order terms is the ***Lack of Fit sum of squares***.

- The ***lack-of-fit test statistic*** is a partial F test for testing the addition of the higher-order terms (up to the highest order) to the polynomial model. Note that the error sum of squares for the highest-order polynomial model is equivalent to the error sum of squares for a model including a dummy variable for each dose level without an intercept (cell means model) or leaving out a reference group if an intercept is included (reference group model). The model sum of squares for the highest order polynomial model and reference group model will also be the same, but not the same as the cell means model (since the cell means model uses the "uncorrected sum of squares").

Lack-of-Fit Test for the straight-line model:

H0: βquad = βcubic = βquartic… = βseptic = 0
H0: $\beta_{X^2} = \beta_{X^3} = \ldots = \beta_{X^7} = 0$

$$F = \frac{(14.372990\text{-}0.2600)/6}{0.01625} = \frac{(169.77958\text{-}155.66669)/6}{0.01625} = \frac{14.1129/6}{0.01625} = 144.75$$

$\sim F_{6,16}$; p < .0001

Lack-of-Fit Test for the quadratic model:

H0: βcubic = βquartic… = βseptic = 0
H0: $\beta_{X^3} = \ldots = \beta_{X^7} = 0$

$$F = \frac{(169.77958\text{-}169.70004)/5}{0.01625} = \frac{0.07954/5}{0.01625} = 0.979$$

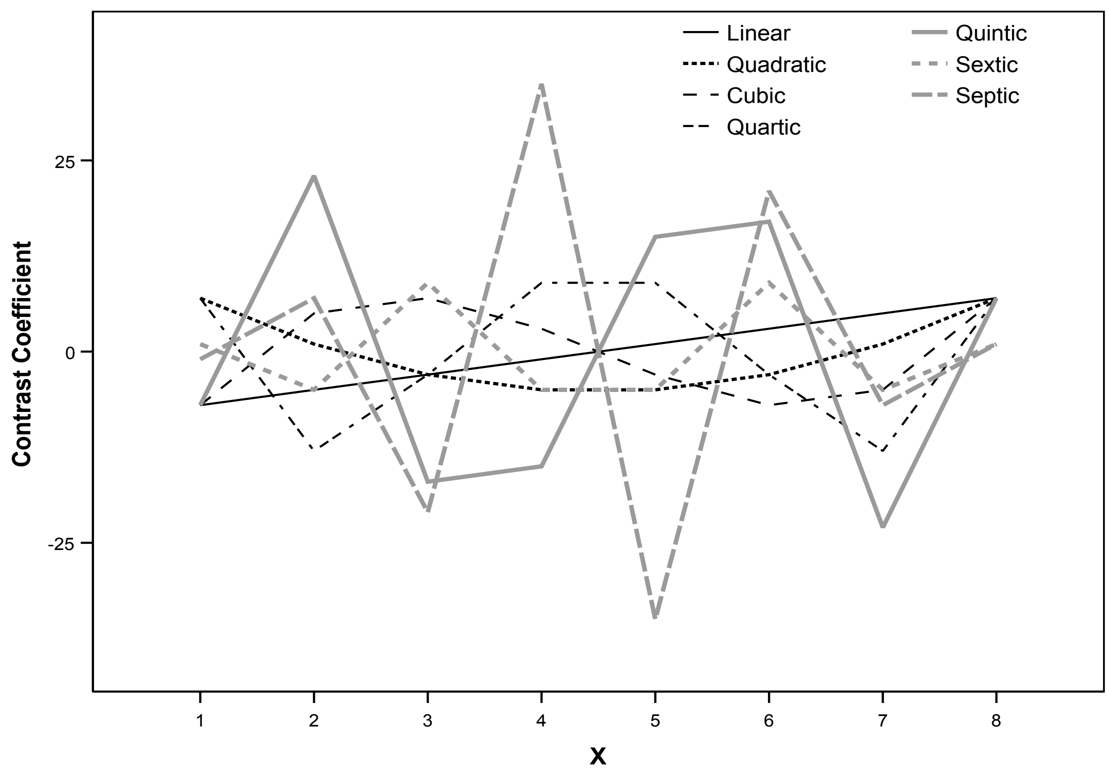
$\sim F_{5,16}$; p = 0.460

Orthogonal Polynomials

- The orthogonal polynomial variables contain exactly the same information as the simple polynomial variables, but unlike the simple polynomial variables, the orthogonal polynomial variables are uncorrelated with each other. These can be used when there are a limited number of value of the predictor (8 levels in the current example).

Coefficients from KKNM Table A7

k=8	X								
	1	2	3	4	5	6	7	8	Σp_i^2
Linear	-7	-5	-3	-1	1	3	5	7	168
Quadratic	7	1	-3	-5	-5	-3	1	7	168
Cubic	-7	5	7	3	-3	-7	-5	7	264
Quartic	7	-13	-3	9	9	-3	-13	7	616
Quintic	-7	23	-17	-15	15	17	-23	7	2184
Sextic	1	-5	9	-5	-5	9	-5	1	264
Septic	-1	7	-21	35	-35	21	-7	1	3432



```
PROC REG DATA=wtgain;
 MODEL wgtgain = odose1 odose2 odose3 odose4 odose5 odose6 odose7;
 LinearLOF: TEST odose2, odose3, odose4, odose5, odose6, odose7;
 QuadLOF:   TEST odose3, odose4, odose5, odose6, odose7;
RUN;
```


Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	169.77958	24.25423	1492.57	<.0001
Error	16	0.26000	**0.01625**		
Corrected Total	23	170.03958			

ANOVA Table Identical to Reference Cell Model, page 40.

Parameter Estimates

	Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
	Intercept	1	3.65417	0.02602	140.43	<.0001
{*Linear*}	ODOSE1	1	**0.55575**	**0.00568**	**97.87**	**<.0001**
{*Quadratic*}	ODOSE2	1	0.16687	0.00568	29.39	<.0001
{*Cubic*}	ODOSE3	1	0.00391	0.00453	0.86	0.4003
{*Quartic*}	ODOSE4	1	-0.00525	0.00297	-1.77	0.0958
{Quintic}	ODOSE5	1	0.00001526	0.00157	0.01	0.9924
{*Sextic*}	ODOSE6	1	-0.00215	0.00453	-0.47	0.6420
{*Septic*}	ODOSE7	1	-0.00112	0.00126	-0.89	0.3871

These parameter estimates can be difficult to interpret. For example, the parameter estimate for ODOSE1 is the expected weight gain for a ½ ounce increase in dose. To obtain weight gain for a 1 ounce increase multiply the parameter estimate by two: 2×0.55575 = 1.1115. However, the p-values tell you whether the addition of each polynomial adds to the model, given the lower order polynomials.

Covariance of Estimates

Variable	Intercept	ODOSE1	ODOSE2	ODOSE3	. . .
Intercept	0.0006770833	0	0	0	
ODOSE1	0	0.0000322421	0	0	
ODOSE2	0	0	0.0000322421	0	
ODOSE3	0	0	0	0.0000205177	
ODOSE4	0	0	0	0	
ODOSE5	0	0	0	0	
ODOSE6	0	0	0	0	
ODOSE7	0	0	0	0	

Test LinearLOF Results for Dependent Variable wgtgain

Source	DF	Mean Square	F Value	Pr > F
Numerator	6	2.35215	144.75	<.0001
Denominator	16	0.01625		

Test QuadLOF Results for Dependent Variable wgtgain

Source	DF	Mean Square	F Value	Pr > F
Numerator	5	0.01591	0.98	0.4602
Denominator	16	0.01625		

Conclusions:

The linear model is not adequate.

A quadratic model is adequate.

To test just the linear effect of dose, we could have modeled the data using a cell means model, and then test the linear contrast (or any other contrast of interest):

```
PROC REG DATA=wtgain;
 MODEL wgtgain = dose1 dose2 dose3 dose4 dose5 dose6 dose7 dose8 / NOINT;
 LINEAR: TEST -7*dose1-5*dose2-3*dose3-1*dose4+1*dose5+3*dose6+5*dose7 +7*dose8;
RUN;
```

```
                         The REG Procedure
                        Dependent Variable: wgtgain

            NOTE: No intercept in model. R-Square is redefined.
```

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	490.25000	61.28125	3771.15	<.0001
Error	16	0.26000	0.01625		
Uncorrected Total	24	490.51000			

This is Σy_i^2 (uncorrected sum of squares) in the cell means model

Pure Error

We are still using the pure error estimate of the MSE, but only need to estimate the contrast of interest (linear contrast).

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
dose1	1	0.86667	0.07360	11.78	<.0001
dose2	1	1.13333	0.07360	15.40	<.0001
dose3	1	1.53333	0.07360	20.83	<.0001
dose4	1	2.20000	0.07360	29.89	<.0001
dose5	1	3.36667	0.07360	45.74	<.0001
dose6	1	4.76667	0.07360	64.77	<.0001
dose7	1	6.66667	0.07360	90.58	<.0001
dose8	1	8.70000	0.07360	118.21	<.0001

This is a cell means model, so each of the beta estimates is the observed mean for that dose group

Test LINEAR Results for Dependent Variable wgtgain

Source	DF	Mean Square	F Value	Pr > F
Numerator	1	155.66669	9579.49	<.0001
Denominator	16	0.01625		

Equivalent to the previous *t* statistics for the linear effect on pages 8 and 14.

F = 9579.49
t = √(9579.49) = 97.875

Appendix 3

Weighted Least Squares (WLS)

- Weighted least-squares can be used when the homogeneity of variance assumption is violated.

- The variances, σ_i^2 for the ith observation, must be known or can be assumed to be of the form $\sigma_i^2 = \sigma^2 / W_i$, where the weights, W_i, are known.

  - With a large number of replicate observations (or by identifying "near neighbors"), the variance can be estimated from the data.

  - The variance can be estimated for each possible value of x_i for a limited number of x_i, or can be a function of X (e.g., if $\text{Var}(\varepsilon_i) = \sigma^2 x_{ij}$, then $w_i = 1/x_{ij}$).

- WLS can also be used when the observed y_i is actually an average of n_i observation at x_i and if all original observations have constant variance σ^2, then the variance of y_i is $\text{Var}(y_i) = \text{Var}(\varepsilon_i) = \sigma^2 / n_i$, and we would choose the weights as $w_i = n_i$.

Example

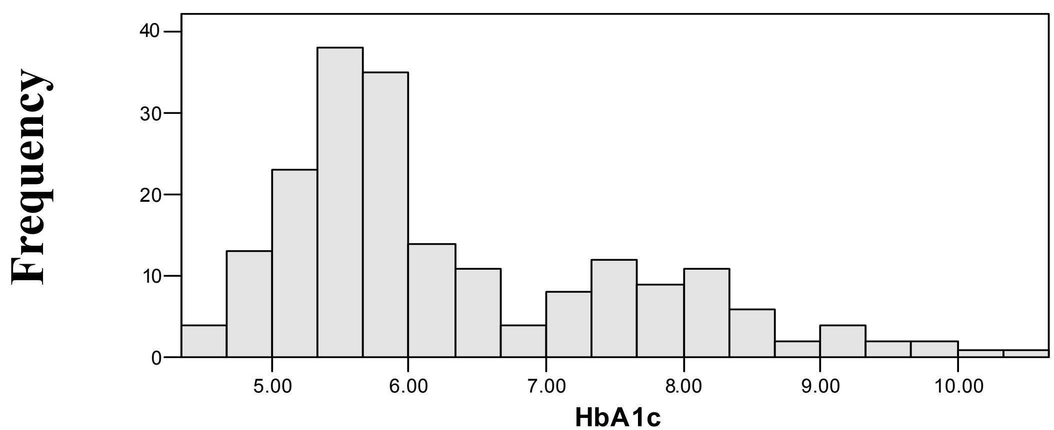
A study was performed to examine whether dietary fiber intake has an effect on HbA1c levels. The HbA1c test (hemoglobin A1c test) is a laboratory test used to estimate average blood glucose levels. Normal HbA1c levels are 4%-6%, but are commonly higher in people with diabetes. Fiber was measured as a continuous variable (grams/day) from a food frequency questionnaire. The study sample consisted of 125 non-diabetic and 75 diabetic individuals, for a total of 200 participants. The following variables are available for the analysis:

hba1c: hemoglobin A1c levels (%)

fiber : dietary fiber intake (grams/day)

diabetes: diabetes status (0 = non-diabetic; 1=diabetic)

Histogram of HbA1c levels:



```
PROC REG;
 MODEL hba1c = fiber diabetes;
RUN;
```

Residual plot strongly suggests heterogeneity of variance.

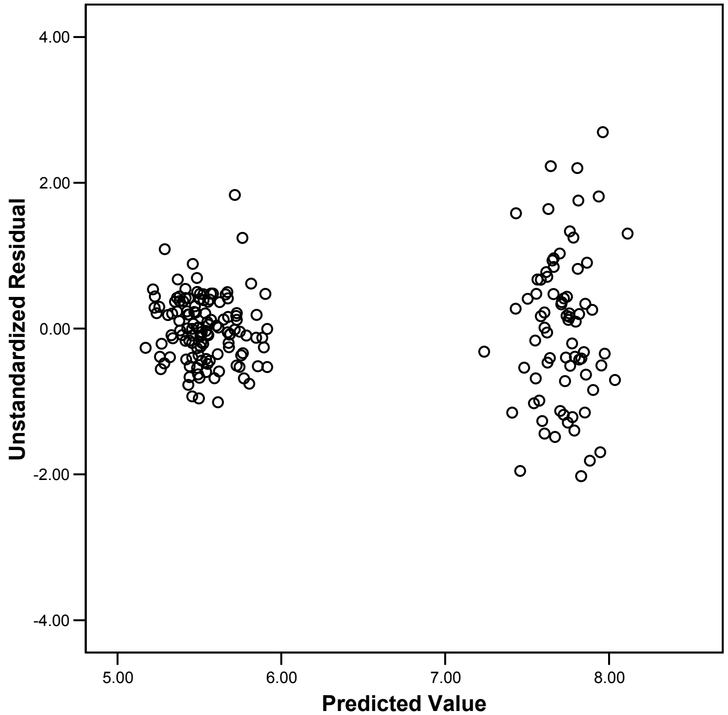
Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	228.84504	114.42252	214.17	<.0001
Error	197	105.25059	0.53427		
Corrected Total	199	334.09564			

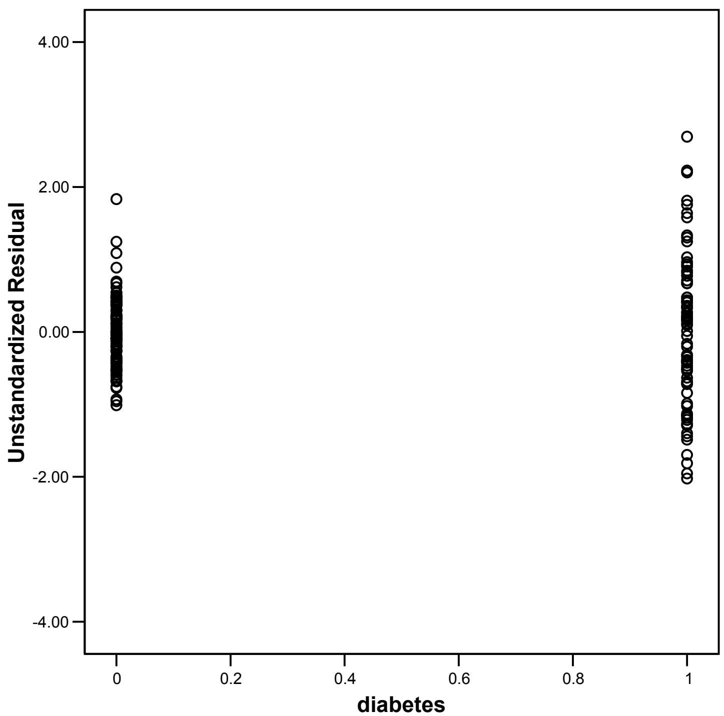
Root MSE	0.73094	R-Square	0.6850
Dependent Mean	6.35115	Adj R-Sq	0.6818
Coeff Var	11.50871		

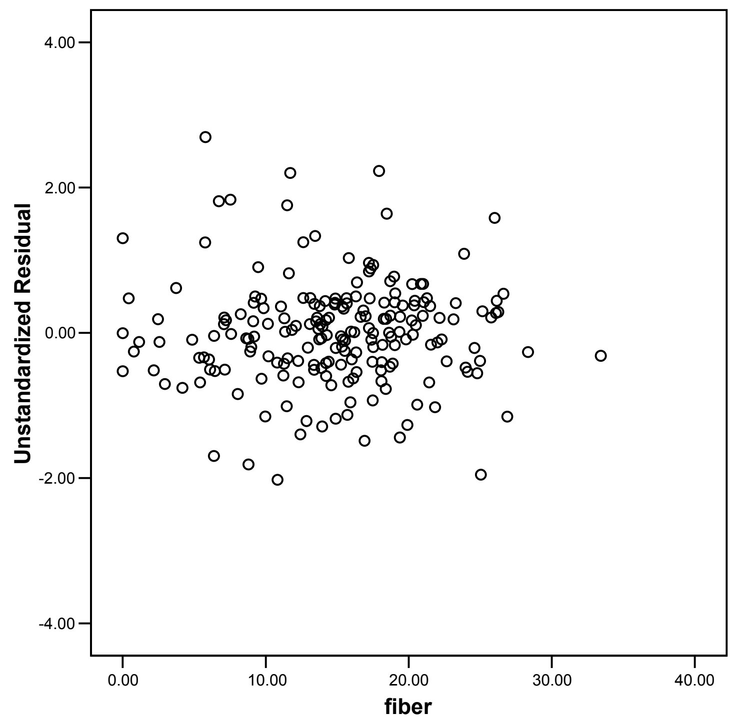
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	5.91423	0.13680	43.23	<.0001
fiber	1	-0.02621	0.00825	-3.18	0.0017
diabetes	1	2.19980	0.10689	20.58	<.0001



Residual plot strongly suggests heterogeneity of variance due to diabetes (due to diabetes).





```
DATA hba1c;
 SET hba1c;
 IF diabetes = 0 THEN vari = 0.4719**2;
 IF diabetes = 1 THEN vari = 1.0592**2;
 weighti = 1/vari;
RUN;

PROC REG DATA=hba1c;
 MODEL hba1c = diabetes fiber;
 WEIGHT weighti;
 OUTPUT out=resids2 predicted=p
residual=r;
RUN;
```

Analysis Variable : hba1c

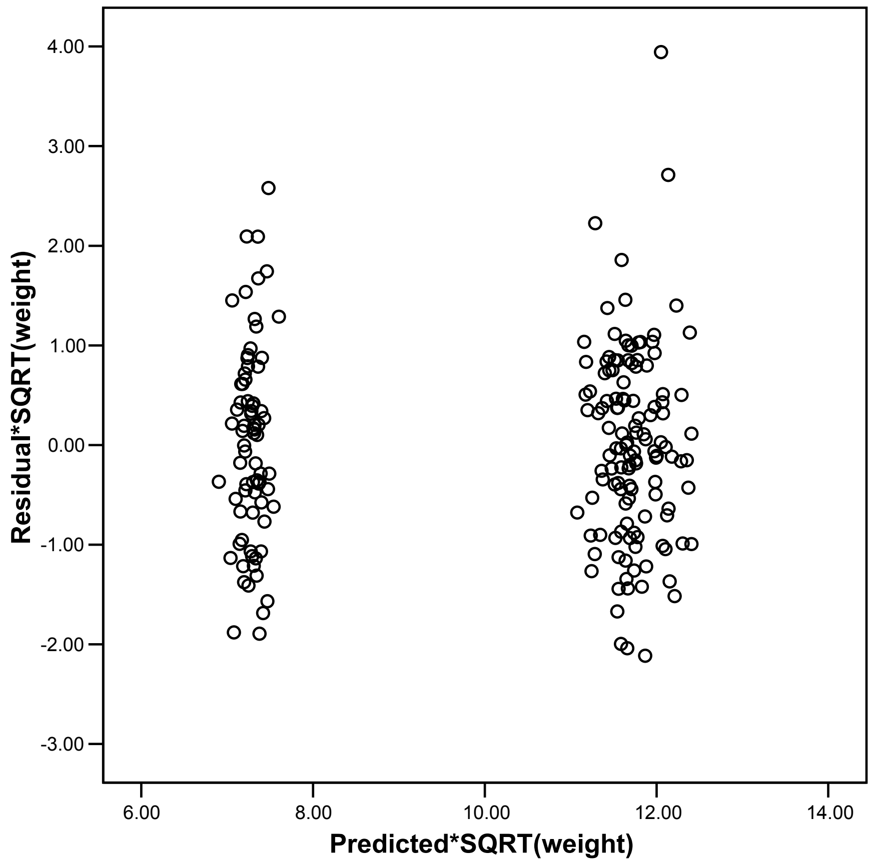
diabetes	N Obs	N	Mean	Std Dev
0=Non-diabetic	125	125	5.5324000	0.4719425
1=Diabetic	75	75	7.7157333	1.0592363

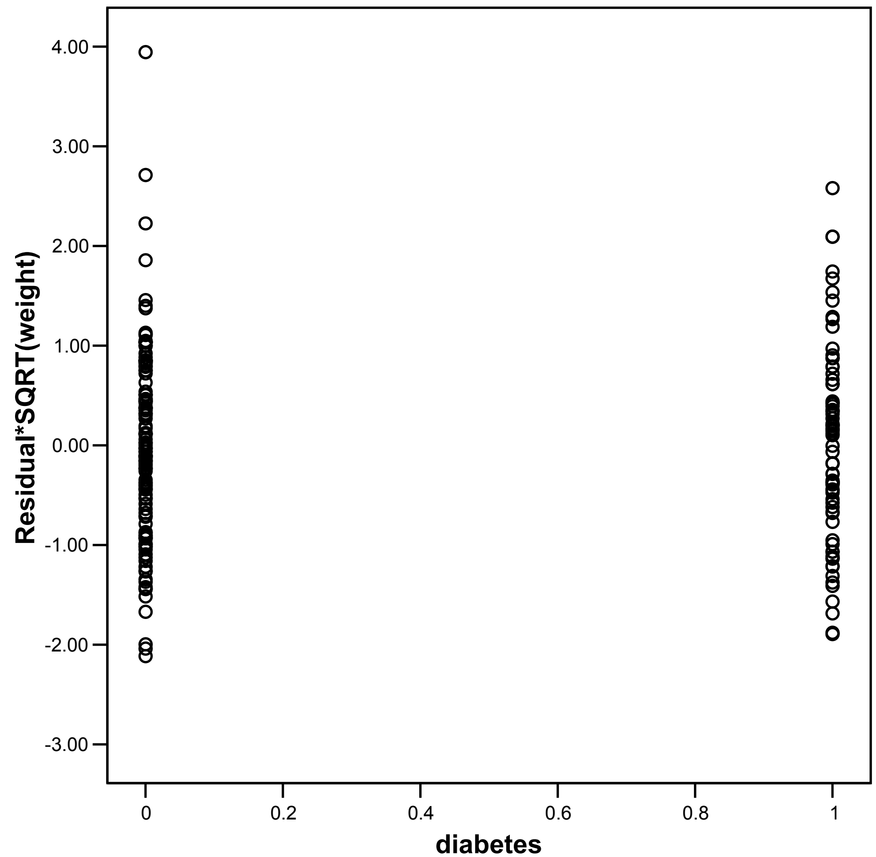
Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	297.39188	148.69594	158.00	<.0001
Error	197	185.39503	0.94109		
Corrected Total	199	482.78692			

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	5.85514	0.09714	60.27	<.0001
fiber	1	-0.02215	0.00605	-3.66	0.0003
diabetes	1	2.19725	0.12557	17.50	<.0001





Residual plots look much improved after fitting weighted least squares.

Same example using diabetes as the only predictor:

```
PROC REG DATA=hba1c;
 MODEL hba1c = diabetes;
 WEIGHT weighti;
RUN;
```

Ordinary Least Squares (OLS):

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	223.45052	223.45052	399.87	<.0001
Error	198	110.64511	0.55881		
Corrected Total	199	334.09564			

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	5.53240	0.06686	82.74	<.0001
diabetes	1	2.18333	0.10918	20.00	<.0001

Weighted Least Squares (WLS):

```
PROC REG DATA=hba1c;
 MODEL hba1c = diabetes;
 WEIGHT weighti;
RUN;
```

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	284.75953	284.75953	284.72	<.0001
Error	198	198.02738	1.00014		
Corrected Total	199	482.78692			

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	5.53240	0.04221	131.07	<.0001
diabetes	1	2.18333	0.12939	16.87	<.0001

Compare to results for two sample t-test:

The TTEST Procedure

Statistics

Variable	diabetes	N	Lower CL Mean	Mean	Upper CL Mean	Lower CL Std Dev	Std Dev
hba1c	0=Non-diabetic	125	5.4489	5.5324	5.6159	0.4198	0.4719
hba1c	1=Diabetic	75	7.472	7.7157	7.9594	0.9127	1.0592
hba1c	Diff (1-2)		-2.399	-2.183	-1.968	0.6806	0.7475

Variable	diabetes	Upper CL Std Dev	Std Err	Minimum	Maximum
hba1c	0=Non-diabetic	0.539	0.0422	4.53	7.55
hba1c	1=Diabetic	1.2623	0.1223	5.5	10.66
hba1c	Diff (1-2)	0.8292	0.1092		

T-Tests

Variable	Method	Variances	DF	t Value	Pr > |t|
hba1c	Pooled	Equal	198	-20.00	<.0001
hba1c	Satterthwaite	Unequal	91.9	-16.87	<.0001

Equality of Variances

Variable	Method	Num DF	Den DF	F Value	Pr > F
hba1c	Folded F	74	124	5.04	<.0001